

IHARQ Cumulative Layer 10 Artifact Package Through P02

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Read-only science boundary. Layer 10 consumes Layer 0 governed claims, the Evidence Map and canonical source data. It does not rerun training, alter populations, select different run subsets, invent tests or strengthen claims.

Executive decision

The complete P00/P01 Layer 10 predecessor package is preserved byte-for-byte. P02 and cumulative artifacts are added as source-data-backed read-only renderings. Every claim-bearing figure/caption is bounded by Layer 0 and mapped through the Evidence Map.

Current artifact inventory

- Figures: **14**
- Tables: **13**
- Deterministic render script: `scripts/render_layer10.py`
- Claim-safe caption registry: `captions/captions.yaml`
- Preserved predecessor Layer 10: `preserved_prior/IHARQ_Cumulative_Layer10_Through_P01_Final(2`

High-value P02 figures

L10-CUM-FIG-001 - Governed phase progression through P02

Claims: CUM-CLM-001/v1, CUM-CLM-002/v1, CUM-CLM-008/v1

Findings: CF-01, CF-02, CF-12

Caption: P00 engineering/foundation contracts enable P01's frozen EEG substrate, which in turn enables P02 Layer-2 decoder evidence. The chain is a provenance/readiness result, not clinical or deployment validation.

Limitations/warnings: TECHNICAL_TRACEABILITY_NOT_EFFECTIVENESS

Output: figures/L10-CUM-FIG-001.png

L10-CUM-FIG-002 - Ablation evidence evolution

Claims: CUM-CLM-005/v1

Findings: CF-05, CF-06

Caption: A4 evolves from P01 matched-substrate feasibility into P02 constrained effectiveness testing; P02 does not establish generic C1-C3 improvement, five-repeat stability, dense-budget characterization or full-grid equivalence.

Limitations/warnings: SINGLE_REPEAT, DEEP_ANCHOR_BUDGETS, POST_HOC_STAGE18S

Output: figures/L10-CUM-FIG-002.png

L10-CUM-FIG-003 - Evidence role evolution

Claims: CUM-CLM-001/v1, CUM-CLM-007/v1

Findings: CF-01, CF-08, CF-09

Caption: Evidence responsibilities progress from foundation validation to data-contract evidence to empirical decoder and negative-result evidence, while phase provenance remains explicit.

Limitations/warnings: EVIDENCE_CLASS_NOT_INTERCHANGEABLE

Output: figures/L10-CUM-FIG-003.png

L10-CUM-FIG-004 - Layer 0 disposition summary

Claims: non-claim governance summary

Findings: none

Caption: Counts of current reviewed claim dispositions. Historical P00/P01 decisions are preserved; P02/cumulative claims are added under the same taxonomy.

Limitations/warnings: COUNT_SUMMARY_ONLY

Output: figures/L10-CUM-FIG-004.png

L10-P02-FIG-001 - P02 A0 full-training performance by dataset

Claims: P02-CLM-002/v1, P02-CLM-003/v1, CUM-CLM-003/v1

Findings: P02-FIND-002, P02-FIND-003, CF-03

Caption: Full-training descriptive ordering differs by dataset. BNCI2014_001 has one held-out test participant; no universal decoder winner is authorized.

Limitations/warnings: BNCI_TEST_PARTICIPANTS_1, DATASET_HETEROGENEITY

Output: figures/L10-P02-FIG-001.png

L10-P02-FIG-002 - Low-label BACC - BNCI2014_001

Claims: P02-CLM-004/v1, CUM-CLM-004/v1

Findings: P02-FIND-005, CF-04

Caption: Descriptive low-label BACC under one frozen subset per budget. Rank changes do not establish repeated-subset sample-efficiency stability.

Limitations/warnings: SINGLE_FROZEN_SUBSET

Output: figures/L10-P02-FIG-002.png

L10-P02-FIG-003 - Low-label BACC - Lee2019_MI

Claims: P02-CLM-004/v1, CUM-CLM-004/v1

Findings: P02-FIND-005, CF-04

Caption: Descriptive low-label BACC under one frozen subset per budget. Rank changes do not establish repeated-subset sample-efficiency stability.

Limitations/warnings: SINGLE_FROZEN_SUBSET

Output: figures/L10-P02-FIG-003.png

L10-P02-FIG-004 - Low-label BACC - PhysioNetMI

Claims: P02-CLM-004/v1, CUM-CLM-004/v1

Findings: P02-FIND-005, CF-04

Caption: Descriptive low-label BACC under one frozen subset per budget. Rank changes do not establish repeated-subset sample-efficiency stability.

Limitations/warnings: SINGLE_FROZEN_SUBSET

Output: figures/L10-P02-FIG-004.png

L10-P02-FIG-005 - Stage 11 diagnostic challenger effect

Claims: P02-CLM-005/v1, P02-CLM-006/v1, CUM-CLM-006/v1

Findings: P02-FIND-006, P02-FIND-007, CF-07

Caption: Diagnostic-only paired delta BACC for the Stage-11 challenger versus canonical primary EEGNet. The challenger is not primary A0 and does not show a consistent cross-dataset improvement.

Limitations/warnings: DIAGNOSTIC_ONLY

Output: figures/L10-P02-FIG-005.png

L10-P02-FIG-006 - A4 C1-C3 role-control effect directions

Claims: P02-CLM-007/v1, CUM-CLM-005/v1

Findings: P02-FIND-009, P02-FIND-010, CF-05

Caption: Governed C1-C3 role-control effects under the accepted constrained A4 scope. Across 756 statistical rows, zero reached raw $p \leq 0.05$; absence of detection is not proof of zero effect.

Limitations/warnings: SINGLE_REPEAT, DEEP_ANCHOR_BUDGETS

Output: figures/L10-P02-FIG-006.png

L10-P02-FIG-007 - Stage18S sign consistency

Claims: P02-CLM-009/v1, CUM-CLM-005/v1

Findings: P02-FIND-012, CF-06

Caption: POST-HOC/DESCRIPTIVE/SENSITIVITY evidence only. Mixed signs reinforce the existing claim ceiling and do not modify canonical Stage18/G18.

Limitations/warnings: POST_HOC_ONLY, THREE_REPEAT_NOT_FIVE_REPEAT

Output: figures/L10-P02-FIG-007.png

L10-P02-FIG-008 - P02 governed non-success categories

Claims: P02-CLM-010/v1, CUM-CLM-007/v1

Findings: P02-FIND-013, CF-08

Caption: FailureCaseIndex categories are governed non-success states and are not interchangeable with unexpected runtime crashes; NegativeResultNotes and DiagnosticOnlyFlags are tracked separately.

Limitations/warnings: FAILURE_CLASSES_NOT_INTERCHANGEABLE

Output: figures/L10-P02-FIG-008.png

L10-P02-FIG-009 - Lee2019_MI participant heterogeneity

Claims: P02-CLM-002/v1, CUM-CLM-003/v1

Findings: P02-FIND-004, CF-11

Caption: Participant-first BACC distributions show substantial heterogeneity; descriptive means do not imply uniform participant benefit.

Limitations/warnings: PARTICIPANT_HETEROGENEITY

Output: figures/L10-P02-FIG-009.png

L10-P02-FIG-010 - PhysioNetMI participant heterogeneity

Claims: P02-CLM-002/v1, CUM-CLM-003/v1

Findings: P02-FIND-004, CF-11

Caption: Participant-first BACC distributions show substantial heterogeneity; descriptive means do not imply uniform participant benefit.

Limitations/warnings: PARTICIPANT_HETEROGENEITY

Output: figures/L10-P02-FIG-010.png

Exact supporting tables

- L10-CUM-TBL-001 - 37 rows - tables/L10-CUM-TBL-001.csv - source table_source_data/L10-CUM-TBL-001.csv.
- L10-CUM-TBL-002 - 3 rows - tables/L10-CUM-TBL-002.csv - source table_source_data/L10-CUM-TBL-002.csv.
- L10-CUM-TBL-003 - 15 rows - tables/L10-CUM-TBL-003.csv - source table_source_data/L10-CUM-TBL-003.csv.
- L10-CUM-TBL-004 - 9 rows - tables/L10-CUM-TBL-004.csv - source table_source_data/L10-CUM-TBL-004.csv.
- L10-P02-TBL-001 - 42 rows - tables/L10-P02-TBL-001.csv - source table_source_data/L10-P02-TBL-001.csv.
- L10-P02-TBL-002 - 72 rows - tables/L10-P02-TBL-002.csv - source table_source_data/L10-P02-TBL-002.csv.
- L10-P02-TBL-003 - 18 rows - tables/L10-P02-TBL-003.csv - source table_source_data/L10-P02-TBL-003.csv.
- L10-P02-TBL-004 - 4 rows - tables/L10-P02-TBL-004.csv - source table_source_data/L10-P02-TBL-004.csv.
- L10-P02-TBL-005 - 3 rows - tables/L10-P02-TBL-005.csv - source table_source_data/L10-P02-TBL-005.csv.
- L10-P02-TBL-006 - 756 rows - tables/L10-P02-TBL-006.csv - source table_source_data/L10-P02-TBL-006.csv.
- L10-P02-TBL-007 - 210 rows - tables/L10-P02-TBL-007.csv - source table_source_data/L10-P02-TBL-007.csv.
- L10-P02-TBL-008 - 45 rows - tables/L10-P02-TBL-008.csv - source table_source_data/L10-P02-TBL-008.csv.
- L10-P02-TBL-009 - 8 rows - tables/L10-P02-TBL-009.csv - source table_source_data/L10-P02-TBL-009.csv.

Reproduction contract

All current figures are regenerated from machine-readable CSV sources by `scripts/render_layer10.py`. Tables are direct copies or deterministic display projections of governed source tables. The script performs plotting/layout only; it does not recompute confirmatory statistics or alter source populations.

Claim safety checks

- P02 A0 ranking figures say **descriptive**, not universal winner.
- Low-label figures state **one frozen subset per budget**.
- Stage11 figure is **diagnostic-only**.
- A4 figure states **0/756 raw $p \leq 0.05$ under the governed constrained scope**, not zero true effect.
- Stage18S figure is visibly **post-hoc/descriptive/sensitivity**.
- Failure figure distinguishes governed non-success classes from unexpected runtime crashes.
- Cumulative progression figures do not imply clinical/deployment validation.

LAYER10_THROUGH_P02_STATUS = FINALIZED